Sorting SOFC plant data into operating stages

Using 50 sensors · how many stages · value for the pattern generator · 2026-10-15 · 39 real days

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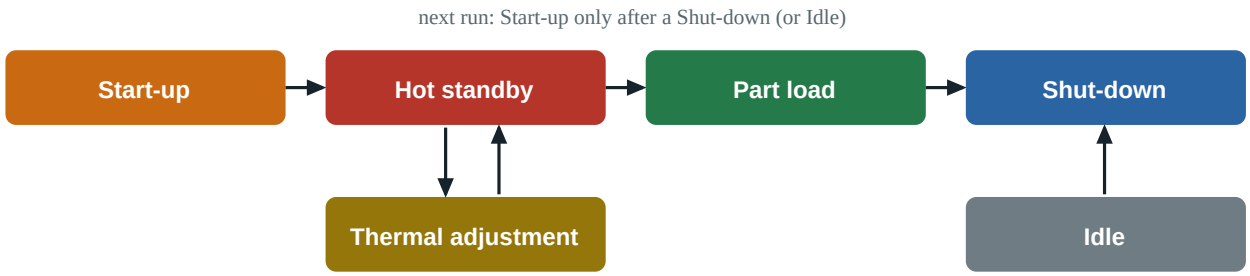
Short answer. Use **6 stages**, plus a 7th (Rated power) that never happened in the data. Sort the minutes with the **rules we already have**: stack temperature and its rate of change, current, heater, air and gas flow, and the order of events. The **50 sensors** are not good enough to set the stages alone: a model trained on them got **86 %** of minutes right on runs it had never seen, and it cannot find Idle. They are very useful as an **independent checker** of the pattern generator's fake data.

1. The six stages



Stage	What it means	Minutes
Shut-down	Cooling down, then cold at room temperature	23 764
Start-up	Heater warms the stack to ~500–600 °C	8 935
Hot standby	Hot, no load, temperature flat	17 839
Thermal adjustment	Hot, no load, temperature moved on purpose	2 746
Part load	Stack delivers current (max 16 A per string)	268
Idle	Warm stack, plant off, no shut-down before	1 258
Rated power	Full design load, ~30 A per string	0

Shut-down has two sub-steps (Cooling, At ambient), so the pattern generator works with 7 building blocks.

If you remember one thing: 6 stages, sorted by rules; the 50 sensors are the second opinion, not the judge.

2. How to sort each minute

Think of a doctor: a few vital signs give the first diagnosis, the patient's history corrects it, and a second doctor double-checks.

Layer	What it does	Where
1 Basic signals	Four yes/no questions per minute: air or gas flowing? current above 1 A? stack above 450 °C? temperature flat (< 5 °C/h)?	operating_states.py
2 Order rules	The stage map fixes look-alikes: Start-up only after Shut-down; Part load only after Hot standby; a warm stack after a Shut-down stays Shut-down.	flow_check.py
3 50-sensor checker	A trained model gives its own guess; disagreements get a second look.	new, tested here

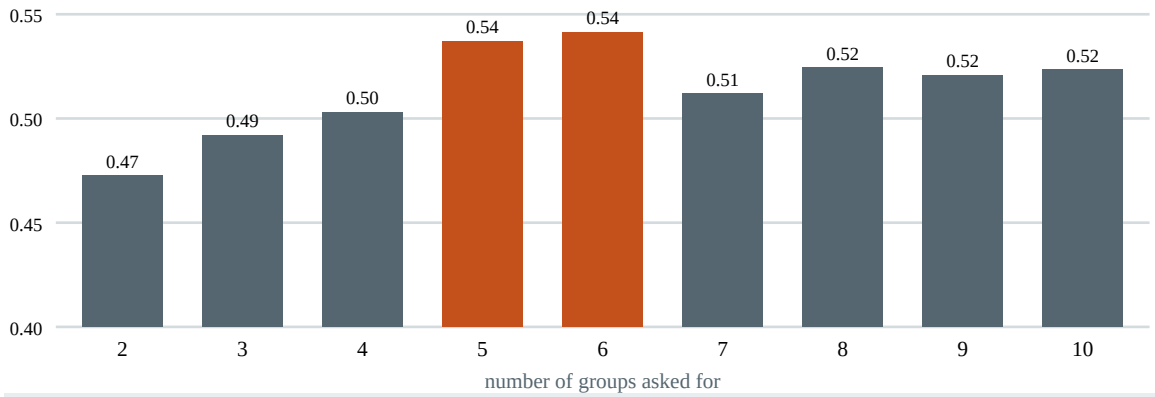
Which 50 sensors: those rated Key or Useful in the traced Excel list that change most between stages; only 6 of the 36 near-identical stack-grid thermocouples kept. Each went in as its value and its 30-minute change.

Kind	Count	Examples
Temperature	26	TIRZ 8,2,5; TIR 4,6,x; TIRC 2,06
Pressure	11	PIR 3,01; PIR 1,4,2; PIR 2,02
Electrical	10	EIRC 4,s,1/2; EIRC 2,01
Flow	3	FIRC 1,4; FIRC 1,3; FIRC 1,5

If you remember one thing: a few signals plus the order of events decide the stage; the 50 sensors double-check.

3. How many stages do the sensors see on their own?

We gave the computer the 50 sensors with no stage names and asked it to group similar minutes (clustering). The groups are most clearly separated at 5 or 6 (silhouette score, higher = cleaner):



Group the data found	Typical minute	Minutes	Our stages inside
Cold, nearly still	67 °C, –2 °C/h	21 246	Shut-down 18 659, Start-up 1 329, Idle 1 258
Hot and steady	511 °C, flat	27 971	Hot standby 17 739, Shut-down 4 989, Start-up 2 835, Thermal adj. 2 407
Heating up	246 °C, +51 °C/h	4 857	Start-up 4 771
Under load	634 °C, 9 A	165	Part load 157
Edge of load	602 °C, ~0 A	269	Part load 108, Hot standby 95, Thermal adj. 64
Hot and rising fast	652 °C, +37 °C/h	302	Thermal adj. 204, Shut-down 97

The data finds physical states (cold, heating, hot and steady, loaded). Our stages also depend on history and intent. Three pairs look alike to the sensors: Idle vs Shut-down (only history differs), Hot standby vs Thermal adjustment (only the direction of change), Start-up vs Shut-down at mid temperature (only the slope).

Recommendation: keep 6 stages (7 with Rated power as a placeholder), Shut-down split into Cooling and At ambient, gas/air phase as an extra column. Agreement between data groups and our stages (adjusted Rand index, 1 = identical): 0.45 at 6 groups, best 0.53 at 8.

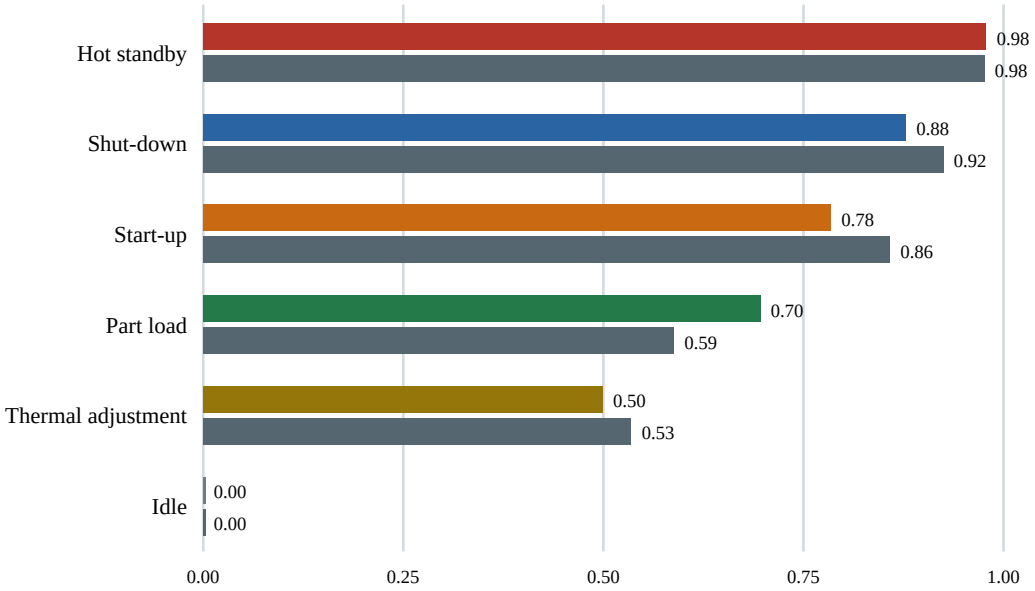
If you remember one thing: the data naturally forms 5–6 groups, close to our 6 stages, but it cannot see history.

4. How well can 50 sensors recognise each stage?

Trained on four groups of days, tested on the fifth, five times over (run folds F1–F5); the model never saw the run it is judged on.

Model input	Minutes right	Avg stage score (F1)
50 sensors, values only	75 %	0.58
50 sensors + 30-min change	86 %	0.64
Same, smoothed 15 min	86 %	0.64
Sensors the rules read (51)	90 %	0.65

Score per stage (coloured = 50 sensors, grey = rule sensors):



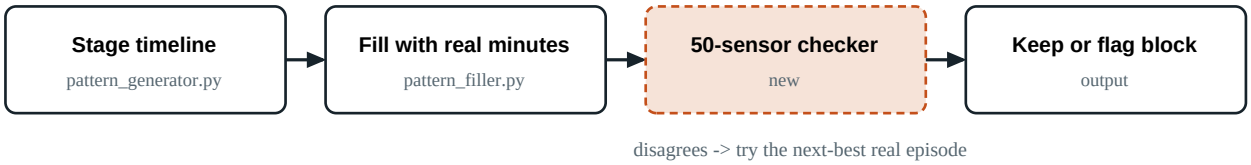
Real \ said	Idle	Start	Hot sb	Th. adj	Part ld	Shut
Idle	0 %	9 %	0 %	0 %	0 %	91 %
Start-up	0 %	83 %	2 %	8 %	0 %	7 %
Hot standby	0 %	0 %	98 %	1 %	0 %	0 %
Thermal adjustment	0 %	9 %	8 %	59 %	0 %	24 %
Part load	0 %	0 %	1 %	21 %	56 %	22 %
Shut-down	0 %	9 %	0 %	4 %	0 %	86 %

Mistakes are the look-alike pairs: Idle called Shut-down (91 %; Idle exists on one day only), Thermal adjustment called Shut-down or Start-up, 9 % of Shut-down called Start-up. More sensors do not add stage information. Per fold: F1 78 %, F2 87 %, F3 96 %, F4 82 %, F5 88 %.

If you remember one thing: 50 sensors get 86 % of minutes right on unseen runs, no better than the rule signals.

5. How useful is this for the pattern generator?

Useful as a quality gate, not as the engine. The generator plans stages and copies real minutes using the rule labels. A 50-sensor model added at the end can check each synthetic block and flag the ones that no longer look like their stage.



Stage	Fake minutes checker agreed with	Verdict
Start-up	98 %	fine
Shut-down	98 %	fine
Part load	97 %	fine
Hot standby	94 %	fine; misses at joins
Thermal adjustment	70 %	weak; inspect
All (seed 7, 12 000 min)	96 %	

Caution: these numbers flatter the generator, because the filler copies real minutes the checker was trained on. The fair accuracy is 86 % on unseen runs.

Helps	Limited	Can't
Flags joins and stretched blocks; double-checks the ~18 % 'outside real range' blocks; one quality number per file	Thermal adjustment (0.50) = warning only; Part load only 268 real minutes	Idle (one day), Rated power (never seen); making stages or curves

Next step: add the checker as an optional last step in pattern_filler.py: per block, the share of minutes it agrees with; below 80 % flag the block or retry with the next-best real episode; skip Idle.

If you remember one thing: use the 50-sensor model as a quality gate on fake data, not to make it.